

MALIGIREDDY JALENDAR REDDY

Senior Data Scientist & Fraud Model Developer • ML Systems • AML/KYC Risk • AWS Production Deployment

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Professional Summary

Senior Data Scientist and Fraud Model Developer with 8+ years of advanced quantitative modeling experience, including 5+ years owning production ML fraud models in fintech and banking. At Fiserv, designed and deployed end-to-end fraud modeling systems — from problem framing through AWS production deployment and ongoing performance management — processing 2M+ transactions/day at sub-300ms latency, preventing ~\$1M+ in annual fraud loss. Expert in Python, SQL, Tableau, statistical modeling (regression, decision trees, gradient boosting, random forests, neural networks, clustering), AUC/KS/precision-recall validation, and drift detection. Proven ability to translate complex model outputs into business impact metrics, influence fraud strategy, and manage model risk at scale. M.S. Data Science (GPA 3.6) · Doctoral Candidate, Information Technology.

Professional Experience

Fiserv | Senior Data Scientist & Fraud Model Developer *May 2024 – Present | New Jersey, USA*

End-to-End Fraud Model Ownership — Problem Framing Through Production

- Owned full fraud model lifecycle across assigned product domains: problem framing → dataset aggregation from Oracle, PostgreSQL, and Snowflake → feature engineering → model training (XGBoost, LightGBM, Gradient Boosting, Random Forests) → validation (AUC, KS, precision/recall, F1) → AWS ECS deployment → ongoing performance monitoring and recalibration — with sole accountability for measurable fraud loss reduction outcomes.
- Designed and deployed scalable, reusable modeling frameworks synthesizing datasets from multiple data environments (Oracle, PostgreSQL, Snowflake, AWS Glue ETL) — reducing model development cycle time ~30% through standardized feature pipelines, documented components, and reusable scoring templates.
- Analyzed complex datasets across Personal Loan, Payment, and KYC/CDD product lines to identify key drivers of fraud loss and member friction; translated analytical findings into actionable model improvements and fraud strategy adjustments presented to senior stakeholders.

Fraud Loss Reduction & False Positive Minimization

- Engineered real-time fraud stop-loss prevention system on Apache Kafka processing 2M+ transactions/day at sub-300ms latency — ML scoring models evaluate every transaction in-flight; automated halt rules prevent suspicious ACH, Wire, Zelle, and Payee transactions before settlement, preventing ~\$1M+ projected annual fraud losses while sustaining <0.5% false positive rate.
- Built and maintained 1,000+ precision-tuned fraud rules across 40+ client portfolios (velocity thresholds, cash patterns, MCC/SIC codes, geo-risk zones, split-transaction detection, counterparty network scoring); managed continuous recalibration against synthetic identity, account takeover, and payment fraud patterns — delivering sustained false positive reductions alongside loss mitigation.
- Conducted trade-off analysis between fraud loss mitigation, customer experience, and regulatory guardrails; influenced risk tolerance decisions and fraud policy for 40+ clients by presenting model performance metrics (AUC/KS trends, false positive rates, loss exposure) to Risk, Fraud Ops, Finance, and Compliance stakeholders.

Model Monitoring, Drift Detection & Risk Management

- Established and maintained model monitoring standards: scheduled AUC/KS drift detection checks, Tableau/Power BI performance dashboards, automated recalibration triggers, and escalation protocols — proactively identified and resolved model decay before business impact across 40+ client configurations.
- Handled escalations related to model performance degradation and risk exposure: diagnosed root causes using CloudWatch, Splunk, and Dynatrace; proposed and implemented recalibration or rule-adjustment solutions; communicated findings and remediation plans to business and compliance stakeholders within SLA.
- Partnered with ML Platform and Engineering teams to productionize models in AWS (ECS, Bedrock, Glue, RDS Aurora); improved model lifecycle governance by defining deployment standards, validation gates, and post-production monitoring checkpoints — reducing time-to-production for new fraud models by ~30%.

External Risk Intelligence, Roadmap Influence & Cross-Functional Partnership

- Investigated external risk data providers (Neustar, LexisNexis, Experian, G2, OFAC) and emerging fraud patterns (synthetic identity, ATO, first-party fraud); translated findings into prioritized roadmap items evaluated by fraud loss ROI, regulatory impact, and implementation effort — directly influencing fraud strategy sequencing.
- Built graph-based network fraud detection layer: traversed account-to-

principal associations, detected split transactions, traced funds across related accounts, and surfaced suspicious counterparty networks using graph database patterns — enriching ML feature sets with structural relationship signals for improved detection.

- Partnered cross-functionally with Risk, Fraud Ops, Engineering, Finance, and Compliance stakeholders to translate complex model outputs into clear business impact metrics (fraud loss prevented, false positive rate, operational cost reduction); independently determined modeling approaches and managed trade-offs with minimal guidance.
- Developed AWS Bedrock AI agent (Claude/Llama) enabling fraud analysts to query alert and case data in natural language — reduced manual case review time ~40%, improved decision traceability, and accelerated escalation resolution across fraud operations teams.

Mukegroup | Software Engineer — AML & Fraud Model Development
Jan 2023 – Apr 2024 | New Jersey, USA

- Designed, validated, and monitored IFM-X transaction monitoring models end-to-end; conducted precision/recall trade-off analysis to minimize false positives while maintaining regulatory coverage; integrated OFAC, LexisNexis, and WLF as real-time external risk signals in decisioning flows.
- Partnered with Risk and Compliance teams on model risk reviews; built Tableau/RCM operational dashboards translating model performance into business metrics; managed P1/P2 production escalations with structured root-cause analysis and documented resolution playbooks.

Tata Consultancy Services | Assistant System Engineer — Fraud & AML Systems *Jan 2020 – Jul 2021 | India*

- Implemented IFM transaction monitoring models with OFAC/WLF external risk data integration; engineered PL/SQL data extraction pipelines for regulatory reporting; automated batch scoring jobs via AutoSys across DEV/UAT/PROD; supported model validation and QA regression cycles.

Technical Skills

ML / Modeling: Python · XGBoost · LightGBM · Gradient Boosting · Random Forests · Decision Trees · Neural Networks · Clustering · Regression · AUC · KS · Precision/Recall · F1 · Drift Detection · Model Recalibration

Data & Visualization: SQL · Snowflake · PostgreSQL · Oracle · AWS Glue ETL · Tableau · Power BI · Pandas · NumPy · Spark

Cloud / Infrastructure: AWS (ECS · Bedrock · Glue · RDS Aurora · Lambda · SQS/SNS · CloudWatch) · Kafka · Redis · Docker · Kubernetes · Jenkins · GitLab CI/CD

Fraud / AML Domain: OFAC · LexisNexis · Neustar · Experian · Graph Fraud Detection · SAR/FinCEN/CTR · KYC · CDD · AML · BSA ·

IFM-X · SAM · HRM · Actimize AIS/RCM · Java 21 · Spring Boot

Education & Certifications

Doctor of Information Technology (Doctoral Candidate) — University of the
Cumberlands *2023 – Present*

M.S. Data Science — University of North Texas | GPA: 3.6 / 4.0 *2021 – 2022*

B.Tech — First Rank · 100% Merit Scholarship *Pre-2020*

*Microsoft Certified: Azure Data Fundamentals • Employee of the Year — Tata
Consultancy Services • IEEE SMC 2026 Peer Reviewer • ConnectSpend iOS
App (App Store) • National Mathematics Olympiad*